

A Project Report on

Automatic Digital Modulation Recognition

Submitted in partial fulfilment of the requirement of
the degree of

Bachelor of Engineering

by

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2024-25



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Certificate

This is to certify that the project entitled '**Automatic Digital Modulation Recognition**' is a bonafide work of

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Project Report Approval for B.E.

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Declaration

We hereby declare that this written submission is a representation of our own ideas, expressed in our own words. Where the ideas or words of others have been included, we have appropriately cited and referenced the original sources. We affirm that we have adhered to the principles of academic honesty and integrity, and that we have not misrepresented, fabricated, or falsified any idea, data, fact, or source within this submission. We understand that any violation of the above principles will result in disciplinary action by the Institute, and may also lead to legal consequences from the sources that have not been properly cited, or from whom permission was not obtained where necessary.

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Submitted with regards,
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List of Abbreviations

ADMR	Automatic Digital Modulation Recognition
SVM	Support Vector Machine
CNN	Convolution Neural Network
GFSK	Gaussian Frequency-Shift Keying
BPSK	Binary Phase-Shift Keying
QAM	Quadrature Amplitude Modulation
AWGN	Additive White Gaussian Noise

Abstract

This project focuses on automatic digital modulation recognition (ADMR) using machine learning and deep learning. A Linear Support Vector Machine (SVM) and a Convolutional Neural Network (CNN) have been implemented to classify digitally modulated signals. The SVM model utilizes manually extracted features, including high-order moments, cumulants, power-based features, spectral features, and their ratios, which help in capturing signal characteristics. The CNN model learns directly from spectrogram representations, employing convolutional layers and max pooling for automatic feature extraction. This enables the CNN to efficiently capture time-frequency patterns. The models have been evaluated using custom dataset and standard dataset. The standard RML.2016.10b dataset, widely used in ADMR research. The dataset consists of GFSK, BPSK, QPSK, 8PSK, 16QAM, and 64QAM, covering different modulation types. Signals have been processed at SNR levels ranging from 0 to 16 dB, ensuring robustness under noisy conditions. These modulation schemes are widely utilized in modern wireless communication systems, including Wi-Fi, LTE, satellite communications, software-defined radios, and cognitive radio networks, ensuring efficient data transmission. This project evaluates the performance of feature-based classification (SVM) and deep learning-based spectrogram analysis (CNN) to determine the best approach for ADMR. SVM achieved an accuracy of 70–80%, while CNN achieved an accuracy of 80–90%.

Chapter 1

Introduction

In the rapidly evolving landscape of wireless communications, the accurate and efficient identification of modulation schemes plays a pivotal role in ensuring reliable signal transmission and effective spectrum monitoring. Automatic Digital Modulation Recognition (ADMR) is a fundamental component in modern intelligent communication systems and finds critical applications in areas such as cognitive radio, electronic surveillance, electronic warfare, and software-defined radio (SDR). As wireless systems become increasingly complex and adaptive, the ability to recognize modulation schemes automatically under varying conditions becomes essential for maintaining communication reliability and situational awareness. This project is focused on developing a robust and scalable ADMR system capable of classifying a wide range of digital modulation techniques, including GFSK, BPSK, QPSK, 8PSK, 64QAM, 16QAM. The system is designed to perform accurately even under challenging channel conditions, such as Additive White Gaussian Noise (AWGN), fading, and interference. A key emphasis is placed on recognizing modulation schemes at low Signal-to-Noise Ratio (SNR) levels, simulating real-world environments where signal degradation is common. To achieve this objective, the project employs a dual-model approach that combines both traditional and deep learning techniques: Support Vector Machine (SVM) and Convolutional Neural Network (CNN). SVM is a powerful supervised learning algorithm known for its effectiveness in classification tasks, particularly when the dataset is limited and features are well-engineered. In this study, statistical features such as higher-order moments, kurtosis, skewness, and spectral properties are extracted from the raw signals and used to train the SVM model for accurate classification. In parallel, a deep learning approach using CNN is implemented to leverage its strength in automated feature extraction and pattern recognition. CNNs are particularly effective when signal data is transformed into time-frequency representations such as spectrograms or scalograms. These 2D visual representations allow the CNN to learn intricate patterns associated with different modulation types, enabling high performance even in noisy or distorted signal environments. This approach reduces the need for manual feature engineering and allows the model to generalize better across different scenarios. By integrating both SVM and CNN methodologies, the project enables a comprehensive evaluation of their respective strengths and limitations. Performance is assessed using High Order Moments (HOMs) and High Order Cumulants (HOCs) to capture signal distribution characteristics. Frequency-domain features using FFT for spectral analysis. Wavelet Transform features to capture transient and multiresolution characteristics.

1.1 Motivation

The motivation for this project arises due to the growing demand for real-time communication systems that require fast and accurate modulation recognition to ensure reliable data transmission. As applications such as autonomous systems, wireless networks, and security surveillance become more advanced, they operate in dynamic environments with fluctuating signals and interference. Automatic Digital Modulation Recognition (ADMR) is crucial for maintaining efficient spectrum usage, minimizing signal interference, and optimizing communication performance. The ability to quickly classify modulation types in real-time is essential for seamless, high-performance operation in these modern communication systems.

1.2 Scope of the Project

The scope of this project is to develop an efficient and robust system for Automatic Digital Modulation Recognition (ADMR) using advanced machine learning and deep learning techniques, specifically Support Vector Machine (SVM) and Convolutional Neural Networks (CNN). The system aims to automatically classify and identify various digital modulation schemes such as GFSK, BPSK, 8PSK, QPSK, 16QAM, 64QAM even under noisy channel conditions with varying Signal-to-Noise Ratio (SNR) levels.

The project will involve:

1. The acquisition or simulation of modulated signals across different modulation types and SNR conditions.
2. Preprocessing and feature extraction, where key statistical or time-frequency characteristics will be derived from the signals for effective classification.
3. Implementing and comparing SVM, a traditional but powerful classifier, with CNN, a deep learning model capable of learning hierarchical features directly from raw signal data or their transformed representations (e.g., spectrograms or I/Q plots).
4. Evaluating the performance of both models using metrics like accuracy, confusion matrix, High Order Moments (HOMs) and High Order Cumulants(HOCs) to capture signal distribution characteristics. Frequency-domain features using FFT for spectral analysis.
5. Studying the impact of varying SNR levels on classification accuracy to assess robustness in real-world conditions.

This project also aims to explore the comparative performance of a basic SVM model versus an advanced architecture, such as a Convolution Neural Network, to demonstrate improvements in modulation recognition accuracy and generalization. By integrating deep learning with conventional machine learning methods, the system aspires to enhance recognition performance, minimize misclassification, and enable practical applications in software-defined radios, cognitive radio systems, and modern communication receivers. Ultimately, the project seeks to develop a reliable, scalable, and automatic recognition system that ensures high identification accuracy of modulation schemes even in challenging environments.

1.3 Problem Statement

To develop an automatic, accurate and efficient digital modulation recognition system that ensures the identification of modulation schemes, focusing on maximizing the identification accuracy even at low levels of signal-to-noise ratio (SNR). The core problem addressed in this project is to design and implement an Automatic Digital Modulation Recognition (ADMR) system that can accurately classify different digital modulation types using advanced machine learning and deep learning techniques. Specifically, the project focuses on utilizing a Support Vector Machine (SVM) classifier and a Convolutional Neural Network (CNN) model to perform modulation recognition based on features extracted from the signal. The goal is to compare the effectiveness of these approaches under varying SNR levels and to assess their classification accuracy and robustness. This project aims to bridge the gap between conventional classification methods and modern learning-based approaches by exploring the performance and limitations of both, ultimately contributing to the development of more adaptive, efficient, and intelligent modulation recognition systems

1.4 Objectives

1. To accurately identify and differentiate between various modulation types, including GFSK, BPSK, QPSK, 8PSK, 64QAM and 16QAM.
2. To analyze and determine the most suitable machine learning or deep learning algorithm for effective modulation classification.
3. To Implement SVM and CNN model.
4. To create a customized dataset and work with standardized datasets.
5. To execute the performance analysis of the model .
6. To conduct a comparative analysis between a Support Vector Machine (basic) model and a deep learning-based Convolutional Neural Network (CNN) model.
7. To obtain the accuracy for varying signal to noise ratio (SNR)

1.5 Outcomes

1. Successfully build a customized dataset with various modulation techniques (GFSK, BPSK, QPSK, 8PSK, 16QAM, 64QAM) at different SNR levels (from 0 to 16 dB).
2. Successfully build both SVM (Support Vector Machine) and CNN (Convolutional Neural Network) machine learning models for modulation classification.
3. The models will be able to classify between GFSK, BPSK, QPSK, 8PSK, 16QAM, and 64QAM modulation schemes at different SNR levels from 0 to 16 dB.
4. The models will achieve an accuracy rate of above 70% for classification tasks across various SNR levels.

Chapter 2

Literature Survey

2.1 Literature Survey

Automatic Modulation Classification (AMC) has been a pivotal technique for identifying modulation formats without explicit overhead, thus enabling adaptive transmissions and cognitive radio systems. Likelihood-based methods, such as the tests introduced in [1], can be near-optimal under fully known channel parameters but tend to be computationally expensive and vulnerable to offsets or rapid channel variations. Accordingly, feature-based AMC has emerged as a more feasible alternative, extracting discriminative statistics—e.g., higher-order cumulants, wavelet transforms, or differential phase factors—and then classifying them via machine-learning models [2, 3]. Empirical evidence in [4, 5] shows that advanced feature engineering and SVM kernel optimization can significantly boost recognition under fading channels. Recent advancements in deep learning have opened new avenues for AMC, particularly under realistic wireless impairments. A CNN-based architecture introduced in [6] utilizes asymmetric convolution blocks and skip connections to extract robust features from raw IQ data, showing high classification accuracy even at low SNRs (e.g., 86.1% at -2 dB). In [7], a dual-branch fusion model integrates both IQ signals and constellation diagrams, using complementary representations to boost classification accuracy while maintaining robustness against noise and frequency offsets. Another advanced approach [8] proposes a hybrid model based on CNNs, transformers, and graph neural networks (GNNs), converting IQ signals into graph structures to extract deeper spatial and temporal features. Similarly, [9] presents a CNN-based model trained on IQ signal representations, demonstrating that converting signals into 2D images enables deep networks to perform modulation recognition effectively without manual feature extraction. Military applications, as discussed in [10], also benefit from these robust AMC approaches under adversarial environments. Although feature-based SVM pipelines remain practical, deep learning has gained prominence for its end-to-end capability and resilience to complex channel impairments. In [11], a CNN–GRU hybrid leverages temporal dependencies for MPSK, MFSK, and MQAM at low SNR, and further approaches incorporate specialized pre-processing—e.g., deep K-SVD denoising in [12] or resolution-transformed spectrograms in [13]. A Deep Belief Network in [14] similarly enhances recognition for ASK and PSK, and [15] demonstrates how CNNs can address phase offsets for OFDM transmissions. Collectively, these works affirm that a judicious blend of domain-specific feature extraction or deep-learning architectures, tailored to channel constraints, can achieve robust AMC performance across diverse applications.

2.2 Applications of Automatic Digital Modulation Recognition

Automatic Digital Modulation Recognition (ADMR) has a wide range of applications across various industries, especially as wireless communication systems become more complex and diverse. Here are the key applications of ADMR:

1. Cognitive Radio (CR) and Dynamic Spectrum Access

- **Spectrum Monitoring:** ADMR is crucial for cognitive radios, which need to identify available frequency bands and adapt to different communication protocols by recognizing the modulation types in use. This allows for more efficient spectrum sharing in crowded environments.
- **Dynamic Spectrum Access (DSA):** Cognitive radios dynamically adjust their transmission parameters based on recognized modulation schemes, helping to avoid interference and optimize network performance.

2. Military and Defense

- **Electronic Warfare:** ADMR is used to detect, classify, and potentially counteract enemy communication systems by recognizing their modulation schemes. This is essential for tasks like signal interception, jamming, and secure communication.
- **Surveillance and Intelligence:** Modulation recognition enables the interception and analysis of signals in hostile environments, identifying the communication protocols of adversaries to gather intelligence.

3. Wireless Communication Networks (5G/6G)

- **Network Optimization:** In modern wireless networks, ADMR helps in optimizing communication protocols by dynamically adapting to changing conditions such as interference, traffic load, and user requirements.
- **Heterogeneous Networks (HetNets):** In 5G and 6G systems, devices often operate across multiple communication technologies (e.g., Wi-Fi, LTE, 5G). ADMR facilitates seamless communication by identifying the modulation schemes used in different technologies.
- **Interference Management:** ADMR can assist in detecting and mitigating interference from other users, especially in dense networks where multiple modulation schemes may coexist in close proximity.

4. Internet of Things (IoT)

- **Device Identification:** ADMR helps in identifying the communication protocols and modulation schemes used by various IoT devices, ensuring proper interoperability and communication in a highly diverse network environment.
- **Energy Efficiency:** In IoT networks, ADMR can optimize power consumption by selecting the most efficient modulation schemes based on environmental conditions and network requirements, enhancing overall energy efficiency.

5. Wireless Sensor Networks (WSNs)

- Adaptive Communication: ADMR enables adaptive communication strategies in WSNs by recognizing the modulation used in different sensor nodes, enhancing data transmission reliability and reducing energy consumption.

6. Remote Sensing and Space Communications

- Satellite Communication: In space communications, recognizing modulation schemes is essential for robust data transmission between satellites and ground stations, especially under varying environmental conditions.
- Remote Sensing Applications: In remote sensing, ADMR can be applied to classify signals received from different sensors or platforms, aiding in the analysis of the collected data.

7. Healthcare

- Remote Patient Monitoring: ADMR ensures reliable wireless transmission of biomedical signals (like ECG and EEG) in wearable and remote monitoring systems, maintaining data accuracy even in noisy environments or areas with low signal strength.
- Medical Imaging and Tele-Surgery: By dynamically identifying and adapting modulation schemes, ADMR supports real-time, high-resolution data transfer needed for applications like robotic surgery and remote diagnostics, reducing transmission errors and latency.

2.3 Literature Survey for Automatic Digital Modulation Recognition

Through our literature survey, it was clear that Support Vector Machines (SVM) consistently offer strong generalization capabilities for Automatic Modulation Classification (AMC) tasks. Zhang et al. [4] demonstrated the effectiveness of the smooth SVM (SSVM) model, but its high computational complexity led us to opt for a simpler SVM model. Based on the literature and performance comparisons, we chose the standard SVM model to balance accuracy and computational efficiency, making it ideal for real-time modulation recognition in varying noise conditions. In parallel, recent developments in deep learning—particularly Convolutional Neural Networks (CNNs)—have significantly advanced AMC performance. CNN-based models such as those proposed in [6, 7] can automatically learn robust features from raw IQ signals or constellation diagrams, offering high classification accuracy even at low signal-to-noise ratios. These models eliminate the need for manual feature extraction and demonstrate resilience to channel impairments. However, their computational demands and requirement for large training datasets present challenges for real-time or embedded deployment. While CNNs represent a powerful solution for complex AMC environments, we found SVM to be a more practical choice under resource-constrained conditions.

Sr. No	Title, Author, and Year of Publication	Modulation Techniques Identified	Methods Used	Feature Extraction Parameters	Key Takeaways	Limitations
1	Automatic digital modulation recognition based on machine learning algorithms by Sam Ansari, Khawla A. Alnajjar, Saeed Abdallah, and Mohamed Saad, 2020	• ASK • QASK • FSK • QFSK • PSK • QPSK	• KNN • PNN	Second-Order Moment of Instantaneous Phase, Spectrum-Based Feature, Mean Power Spectral Density, Standard Deviation of Non-Linear Phase Component, Two-Dimensional Wavelet Transform Correlation	Significant performance improvements over existing methods; KNN provided up to 88.5% faster execution.	Focuses on limited modulation types and does not explore deep learning models.
2	Automatic digital modulation recognition based on novel features and support vector machine by Salman Hassanpour, Amir Mansour Pezeshk, and Fereidoon Behnia, 2016	• BASK • BFSK • BPSK • 4-ASK • 4-FSK • QPSK • 16-QAM	Support vector machine (SVM)	Temporal and spectral-based features, wavelet domain features	High accuracy in low SNR conditions using SVM.	Limited to specific modulation types, reducing generalization to other schemes.
3	Automatic Modulation Classifier for Military Applications by Víctor Iglesias, Jesús Grajal, Omar Yeste-Ojeda, 2011	• LFM • NFM • PSK (2PSK, 4PSK, MPSK)	Likelihood-Based (LB) Methods, Feature-Based (FB) Methods, Block-by-Block Scheme	Mean Square Error (MSE) of Unwrapped Phase, Variance of Instantaneous Frequency	Improves latency and real-time operation for military applications.	Does not consider performance under low SNR or noisy conditions.
4	Automatic Modulation Classification Using a Support Vector Machine-Based Pattern Recognition Algorithm by Varna Kumar Reddy P. G., Meena M., 2022	• BPSK • QPSK • 8PSK • 4QAM • 16QAM • 64QAM	SVM (Linear, Quadratic, Cubic, Gaussian)	Statistical features under non-ideal channel conditions, cumulants, HOC ratios	Polynomial kernel outperforms other kernels and traditional methods; effective with less training data and lower SNR.	Lacks exploration of deep learning alternatives; limited performance at extremely low SNR.

Table 2.1: Literature Survey - Part 1

Sr. No	Title, Author, and Year of Publication	Modulation Techniques Identified	Methods Used	Feature Extraction Parameters	Key Takeaways	Limitations
5	Automatic Modulation Recognition Based on CNN and GRU by Fugang Liu, Ziwei Zhang, and Ruolin Zhou, 2022	• 2ASK • 2PSK • 2FSK • 4ASK • 4PSK • 4FSK • 16QAM • 64QAM	CNN, GRU, CNN+GRU	HOC characteristics, Wavelet transform, Instantaneous characteristics, Cyclic spectrum characteristics	CNN+GRU achieves more than 90% accuracy at SNR = -6 dB, 100% at SNR = -1 dB.	High computational cost; lacks generalization to other modulation techniques beyond those tested.
6	Modulation Recognition of Digital Signals Based on Deep Belief Network by Guo Yunxin, Zhang Yue, Ma Hong, 2019	• 4ASK • BPSK • QPSK • 2FSK • 4FSK	Deep Belief Network	High-order cumulants	Recognition rate of more than 93% for five signals under various conditions.	DBN requires large amounts of training data and high computational resources.
7	Automatic Modulation Classification Based on Novel Feature Extraction Algorithms by Xiaolin Zhang, Jianting Sun, Xiaotong Zhang, 2020	• MSK • 2ASK • 4ASK • 8ASK • 2PSK • 4PSK • 8PSK • 16QAM • 32QAM • 64QAM • 2FSK	Multi-layer smooth SVM classifier with feature selection (FS,DT-SSVM)	Differential nonlinear phase peak factor (PF), cyclic spectrum, power spectral density, HOC features	PF has a significant effect on MSK and 2FSK signal classification when SNR $\geq$ 8 dB.	Reduced performance at SNR $\leq$ 8 dB; lacks evaluation under diverse noise conditions.

Table 2.2: Literature Survey - Part 2

Chapter 3

Methodology

In this project, two models namely, SVM(support vector machine) and CNN (convolution neural network) have been used. For both the models two datasets have been used. One is the standard dataset which is RML2016.10b and the other is a customized dataset. The standard dataset contains 1000 samples for each modulation at each of the values of SNR ranging from 0-16dB. The customized dataset also contains 1000 samples for each of the modulation techniques at each of the values of SNR. From both these datasets 80% dataset is used for training purpose and 20% dataset is used for testing purpose.

3.1 System Model for SVM

The proposed system for Automatic Digital Modulation Recognition (ADMR) consists of two primary phases: the Training Phase and the Testing Phase. Each phase includes a series of signal processing and machine learning steps designed to enable the accurate classification of modulation schemes under noisy conditions.

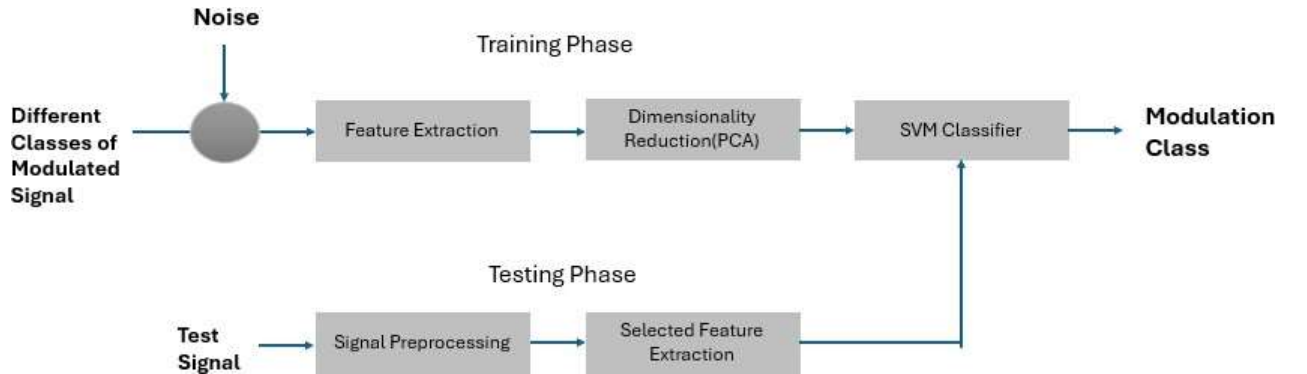


Figure 3.1: Block Diagram for SVM

1. **Input: Different Classes of Modulated Signal + Noise:** In the training phase, the system is fed with signals modulated using various digital modulation schemes (e.g., GFSK, BPSK, QPSK, 8PSK, 16QAM, 64QAM). These signals are deliberately mixed with additive noise to simulate real-world conditions and test the model's robustness. The noise may vary according to different Signal-to-Noise Ratio (SNR) levels to ensure the system learns to generalize under both clean and degraded conditions.

Training Phase

2. **Feature Extraction:** Feature extraction plays a crucial role in accurately distinguishing between different modulation schemes. Common techniques include higher-order statistical moments (mean, variance, skewness, kurtosis), which capture the distribution characteristics of signals. Spectral features like power spectral density and bandwidth help differentiate between narrowband and wideband modulations. Features such as higher-order cumulants provide robustness in noisy conditions, especially useful at low SNRs. Wavelet Transform features to capture transient and multiresolution characteristics.
3. **Dimensionality Reduction:** Once features are extracted, they are often high-dimensional. To reduce computational complexity and improve classification performance, Principal Component Analysis (PCA) is used. PCA transforms the original feature set into a lower-dimensional space while retaining most of the variance, thus preserving essential information while eliminating redundancy and noise.
4. **SVM Classifiers:** We employ Support Vector Machines (SVM) as our primary classification algorithm. SVM is chosen for its effectiveness in handling high-dimensional data and its ability to provide robust classification even in the presence of noise. Various kernel functions are tested to optimize classification performance.

Testing Phase

5. **Input: Test Signal:** During the testing phase, a new or unknown modulated signal is provided as input to the system. This signal may also contain noise and is used to evaluate the classifier's ability to recognize the modulation scheme it was not explicitly trained on.
6. **Signal Preprocessing:** The test signal undergoes preprocessing steps similar to the training signals. This may include normalization, filtering, or transformation into a consistent format suitable for feature extraction.
7. **Selected Feature Extraction:** Only the features identified as most effective during training are extracted from the test signal. This ensures consistency and prevents the use of irrelevant or redundant features that may harm classification accuracy.
8. **Classification (Using Trained SVM Model):** The extracted features are fed into the previously trained SVM classifier. Based on the learned decision boundaries, the classifier predicts the most likely modulation class of the test signal.
9. **Output: Modulation Class:** Finally, the system outputs the predicted modulation class (e.g., GFSK, BPSK, QPSK, 8PSK, 16QAM, 64QAM) completing the recognition process.

3.2 System Model for CNN

The proposed system employs a deep learning-based approach using Convolutional Neural Networks (CNN) to perform Automatic Digital Modulation Recognition (ADMR). This system is structured into two main phases: the Training Phase and the Testing Phase, both designed to process modulated signals and classify their modulation types even in the presence of noise.

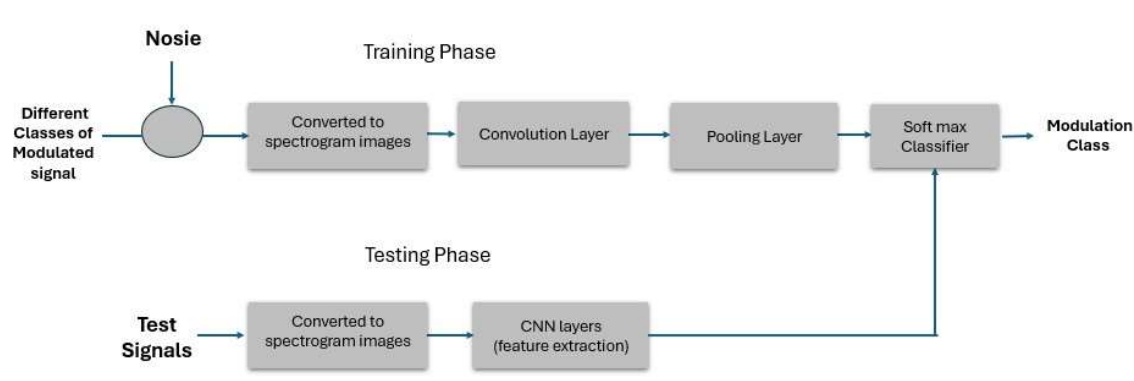


Figure 3.2: Block Diagram for CNN

Training Phase

Input: Different Classes of Modulated Signal + Noise: In this phase, digitally modulated signals are used as input. Noise is intentionally added to these signals to mimic realistic communication scenarios and improve the generalizability of the model.

Converted to Spectrogram Images: Noise modulated signals in the time domain are converted to spectrogram images using techniques such as the Short-Time Fourier Transform (STFT). A spectrogram is a 2D visual representation of the signal's frequency spectrum over time, effectively capturing both temporal and frequency domain features, making it suitable for image-based classification using CNNs.

Convolution Layer: In this layer, multiple filters (kernels) slide over the spectrogram image to extract local patterns such as edges, textures, and shapes that are indicative of specific modulation types. These learned features help differentiate between modulation schemes based on their spectral characteristics.

Pooling Layer: The pooling layer performs dimensionality reduction by summarizing the outputs of the convolutional layers (typically using max or average pooling). This reduces computational complexity, provides translation invariance, and helps prevent overfitting.

Softmax Classifier: The final output of the CNN (after flattening the feature maps) is passed through a Softmax classifier, which produces a probability distribution over the modulation classes. The class with the highest probability is chosen as the predicted modulation scheme.

Testing Phase

Input: Test Signals: New modulated signals are introduced during testing to evaluate the model's performance. These signals are different from the training set and simulate real-world signals the system might encounter.

Converted to Spectrogram Images: Similar to the training phase, the test signals are converted into spectrogram images. This ensures consistency in input format and enables the CNN to use the same learned features for classification.

CNN Layers (Feature Extraction): The trained CNN model processes the spectrogram image through its convolutional and pooling layers. These layers extract hierarchical features which are then used for classification.

Output: Modulation Class: The features extracted by the CNN are passed through the softmax classifier (from the training phase), which outputs the predicted modulation class (e.g., BPSK, QPSK, 16QAM, etc.).

Chapter 4

Implementation and Results

In this project, two datasets have been used. One is the standard dataset which is RML2016.10b and the other is a customized dataset. The standard dataset contains 1000 samples for each modulation at each of the values of SNR ranging from 0-16dB. The customized dataset also contains 1000 samples for each of the modulation techniques at each of the values of SNR. From both these datasets 80% dataset is used for training purpose and 20% dataset is used for testing purpose. Two models SVM(support vector machine) and CNN(convolution neural network) have been used on this datasets. Confusion matrix and classification report with recall , F-1 score, precision, support parameters have been created to evaluate both the models.

4.1 RESULTS(SVM Model)

STANDARDIZED DATASETS

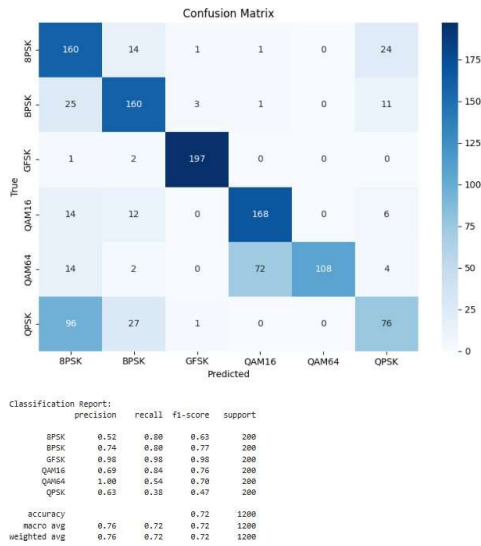


Figure 4.1: SNR = 0 dB

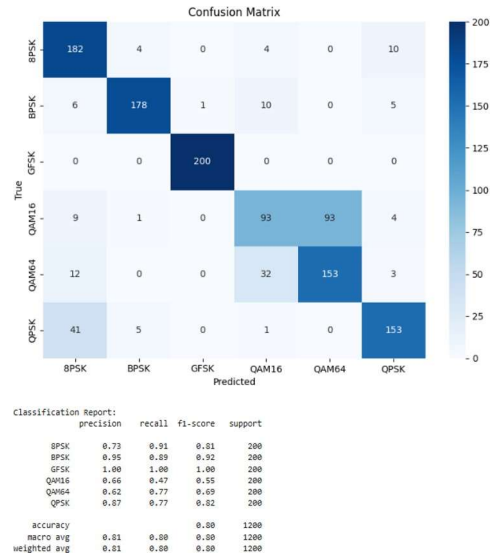


Figure 4.2: SNR = 4 dB

In Figure 4.1 GFSK achieves the highest precision and recall, while QPSK shows significant confusion with 8PSK and BPSK, leading to lower scores. Overall accuracy is 72%, with macro and weighted F1-scores also at 0.77, indicating moderate but imbalanced performance across classes.

In figure 4.2 shows a CNN model achieving 80% accuracy with strong performance on GFSK, BPSK, and QPSK, but moderate confusion between QAM16 and QAM64. The macro and

weighted averages for F1-score and recall are both 0.80, reflecting a strong and fairly balanced performance across most classes.

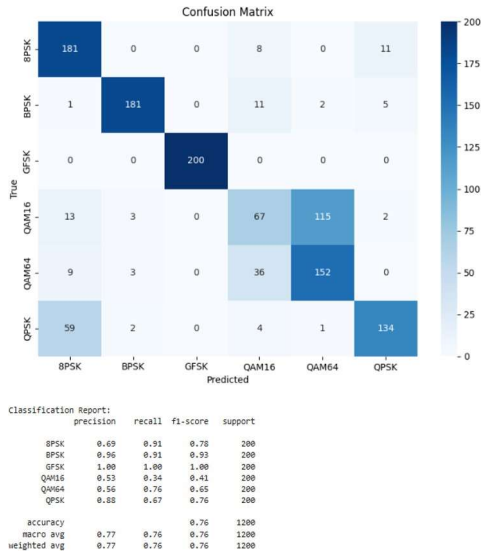


Figure 4.3: SNR = 8 dB

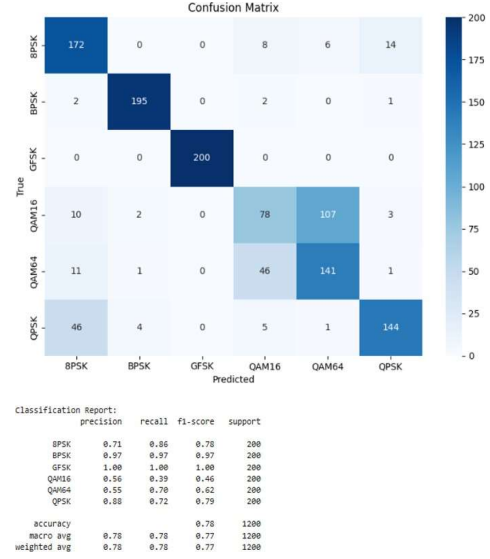


Figure 4.4: SNR = 12 dB

The figure 4.3 the model shows a CNN model with 76% accuracy. GFSK is perfectly classified, while QAM16 is often confused with QAM64, leading to a low F1-score of 0.41. QPSK also shows misclassification with 8PSK, indicating the model struggles with similar modulation types.

The figure 4.4 the model achieves 78% overall accuracy, with strong performance on GFSK (F1-score 1.00) and BPSK (0.97). QAM16 shows the weakest performance with an F1-score of 0.46 due to frequent confusion with QAM64. Most other modulation types are classified accurately with high precision and recall.

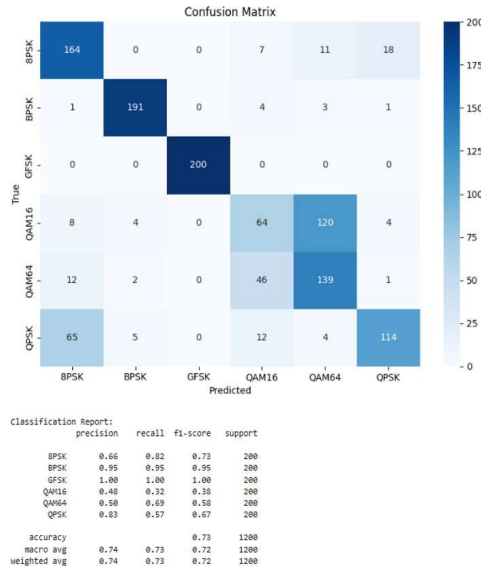


Figure 4.5: SNR = 16 dB

The Figure 4.5 shows a confusion matrix and classification report for a model classifying six modulation types. GFSK and BPSK perform very well, with GFSK achieving perfect accuracy. In contrast, QAM16 and QAM64 show poor performance, with QAM16 having the lowest F1-score. Overall model accuracy is 73%, with an average F1-score of 0.72.

For SVM model with the standard dataset, It is observed that as SNR increases from 0 to 16 dB with values of 0,4,8,12 and 16 dB the model gives an accuracy of 72%, 80%, 76%, 78% and 73% respectively thus averaging to a value of 76% approximately.

CUSTOMIZED DATASETS(SVM)

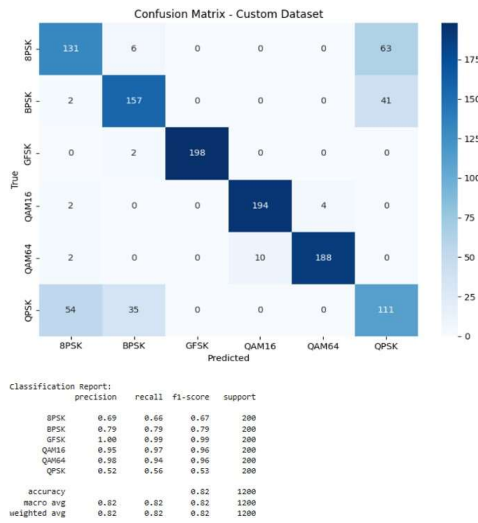


Figure 4.6: SNR = 0 dB

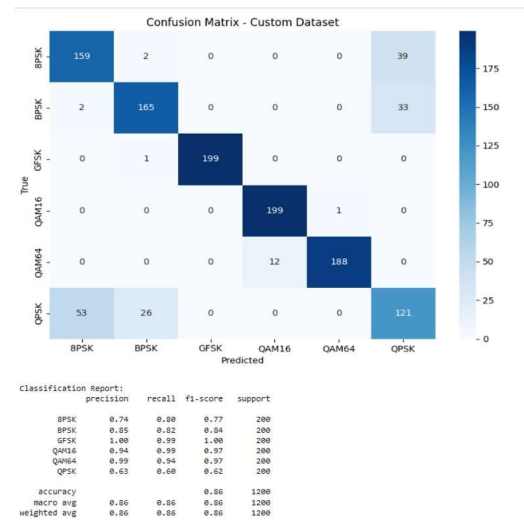


Figure 4.7: SNR = 4 dB

In figure 4.6 the model achieves 82% accuracy with strong performance on GFSK, QAM16, and QAM64, each with F1-scores above 0.95. However, QPSK is frequently misclassified as 8PSK or BPSK, resulting in a low F1-score of 0.53. 8PSK and BPSK also show moderate confusion with QPSK. Overall, the model performs well but struggles mainly with distinguishing QPSK.

In figure 4.7 the model shows improved performance with an accuracy of 86% and high F1-scores for all classes, especially QAM16 and QAM64 (above 0.95). GFSK is nearly perfectly classified, and confusion among classes is notably reduced. QPSK remains the weakest but improves to an F1-score of 0.62. Overall, this model performs better and more consistently across all classes.

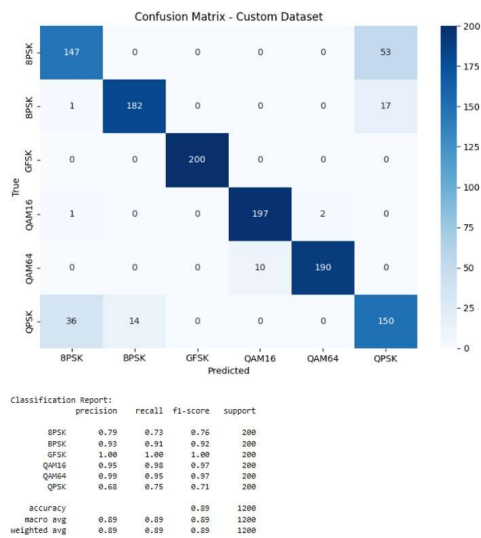


Figure 4.8: SNR = 8 dB

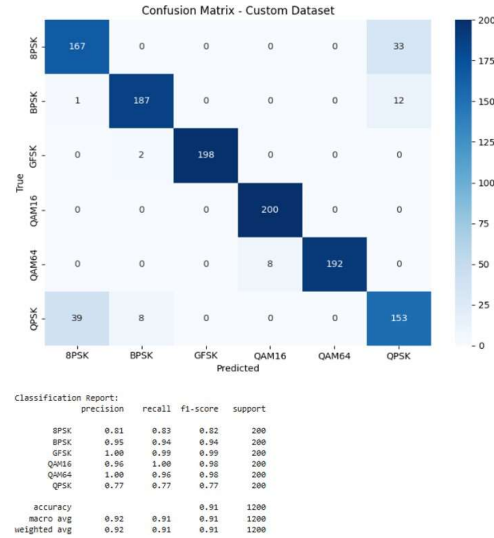


Figure 4.9: SNR = 12 dB

In figure 4.8 the model shows a confusion matrix and classification report for a model classifying modulation types. It achieved 89% accuracy, with GFSK classified perfectly. However, 8PSK and QPSK had notable misclassifications, especially 8PSK often predicted as QPSK. QPSK had the lowest F1-score (0.71), indicating more errors.

In figure 4.9 the model shows a confusion matrix and classification report for modulation classification. The model achieved 91% accuracy, improving over the previous version. GFSK and QAM16 had near-perfect classification with F1-scores of 0.99 and 0.98 respectively. QPSK again had the lowest performance with an F1-score of 0.77.

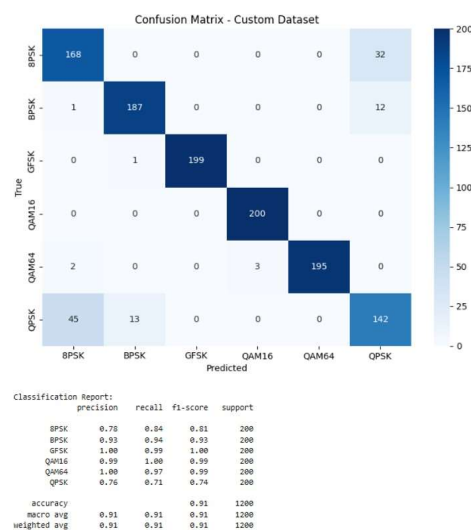


Figure 4.10: SNR = 16 dB

In figure 4.10 the model shows a confusion matrix and classification report for a modulation classification model. The model achieved 91% accuracy, with GFSK and QAM16 performing best (F1-score: 1.00 and 0.99). 8PSK and QPSK had more errors, with QPSK being misclassified mostly as 8PSK (45 times). QPSK again had the lowest F1-score at 0.74, indicating room for improvement.

For SVM model with the customized dataset, It is observed that as SNR increases from 0 to 16 dB with values of 0,4,8,12 and 16 dB the model gives an accuracy of 82%, 86%, 89%, 91% and 91% respectively thus averaging to a value of 88% approximately.

The results for SVM model shows that SVM with customized dataset gave better results with an average accuracy of 88% as against the average accuracy of 76% for the standard dataset.

4.2 RESULTS(CNN Model)

STANDARDIZED DATASET(CNN)

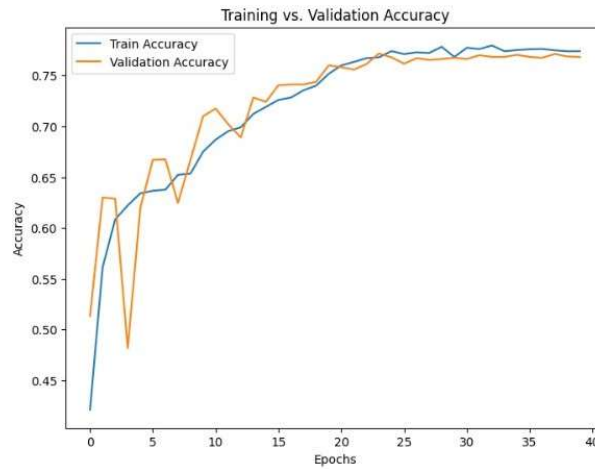


Figure 4.11: Training Vs Validation Accuracy for Standardized Dataset

The graph shows the training and validation accuracy over 40 epochs. Initially, both accuracies increase, with some fluctuations in validation accuracy. As training progresses, both curves rise and converge, indicating the model is learning well. After around 25 epochs, the curves stabilize near 77–78% accuracy. The close alignment of both curves suggests minimal overfitting and good generalization to the validation data.

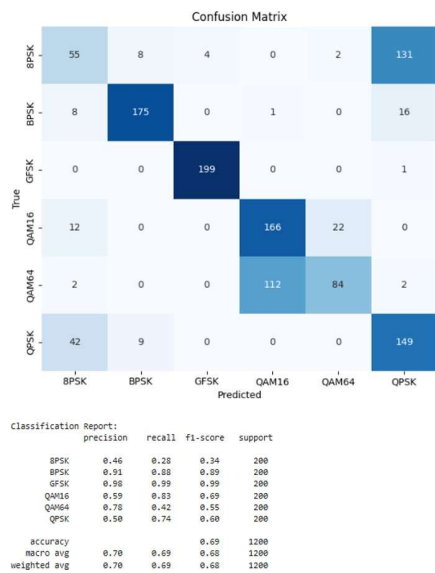


Figure 4.12: SNR = 0 dB

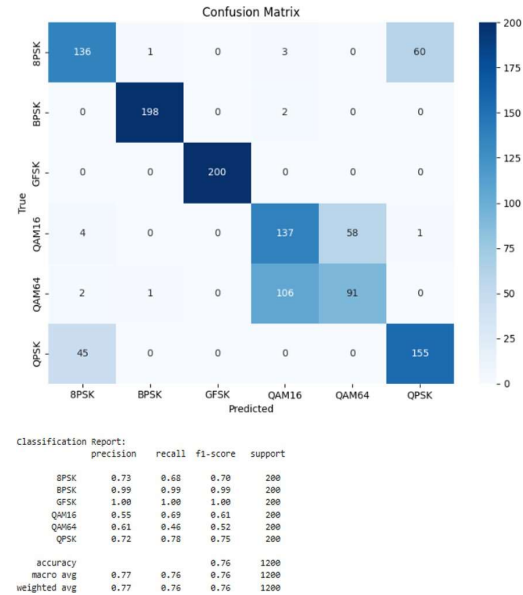


Figure 4.13: SNR = 4 dB

In figure 4.12 the model shows a drop in model performance with overall accuracy at 69%. BPSK was poorly classified with an F1-score of 0.34, heavily misclassified as QPSK. GFSK remained highly accurate with an F1-score of 0.99, while QAM64 and QAM16 saw confusion between each other. QPSK still performed relatively better (F1-score 0.60), but overall results indicate significant classification issues.

In figure 4.13 the model shows a 76% accuracy, performing well for BPSK, GFSK, and QPSK. QAM16 and QAM64 were frequently confused, especially QAM16 misclassified as QAM64 (58 times). GFSK achieved perfect classification (F1-score 1.00), while QAM64 had the weakest F1-score (0.52).

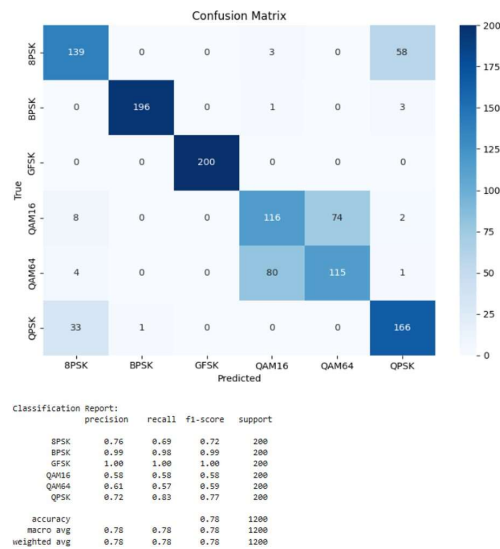


Figure 4.14: SNR = 8 dB

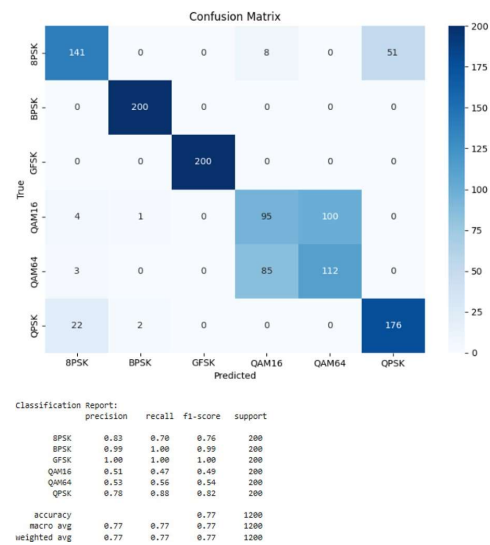


Figure 4.15: SNR = 12 dB

In Figure 4.14 the model achieves 78% accuracy, performing excellently on BPSK, GFSK, and QPSK. QAM16 and QAM64 are often confused with each other, lowering their F1-scores to 0.58 and 0.59. GFSK shows perfect classification, while QPSK is the most accurately predicted among higher-order modulations.

In figure 4.15 This model shows 77% overall accuracy, with perfect predictions for BPSK and GFSK. QAM16 and QAM64 have significant confusion between each other, pulling their F1-scores below 0.55. QPSK achieves strong performance with an F1-score of 0.82, indicating reliable classification.

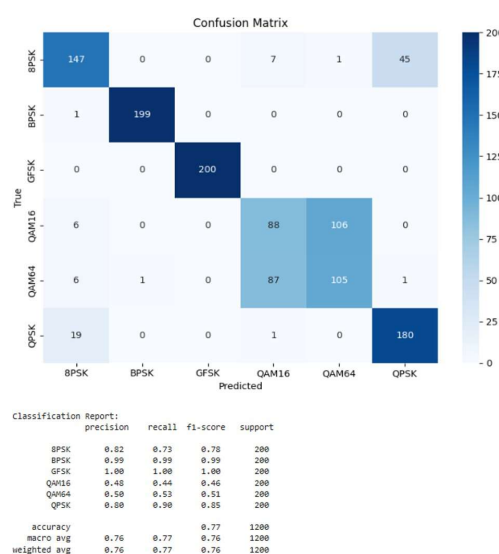


Figure 4.16: SNR = 16 dB

In figure 4.16 The model reaches 77% accuracy, with excellent results for BPSK, GFSK, and QPSK—each achieving F1-scores near or above 0.90. QAM16 and QAM64 continue to show

confusion with each other, lowering their precision and recall. Despite this, overall performance remains balanced with strong macro and weighted averages of 0.76.

For CNN model with the standard dataset, It is observed that as SNR increases from 0 to 16 dB with values of 0,4,8,12 and 16 dB the model gives an accuracy of 69%, 76%, 78%, 77% and 77% respectively thus averaging to a value of 75% approximately.

CUSTOMIZED DATASET(CNN)

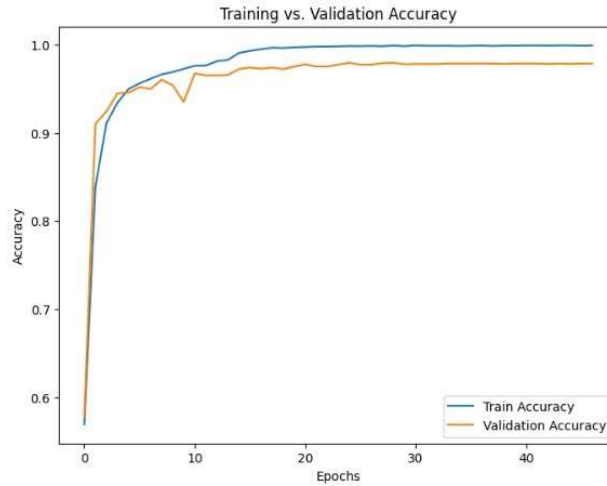


Figure 4.17: Training Vs Validation Accuracy for Customized Dataset

This graph displays training and validation accuracy over 45 epochs. Both accuracies rapidly increase in the first few epochs, indicating fast initial learning. The training accuracy reaches nearly 100%, while validation accuracy plateaus just below that. The small gap between the two suggests slight overfitting, but overall good performance. The model shows strong learning ability and generalizes well on unseen data.

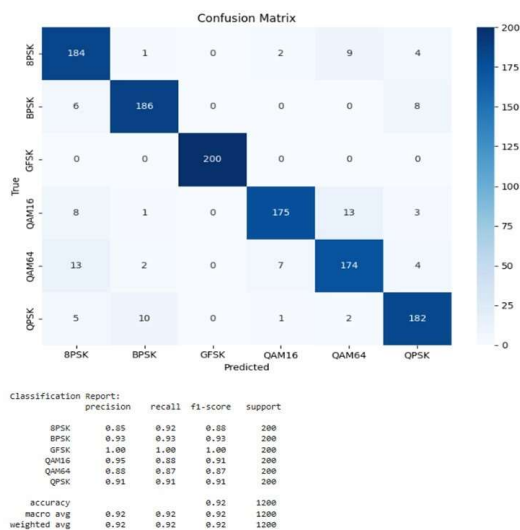


Figure 4.18: SNR = 0 dB

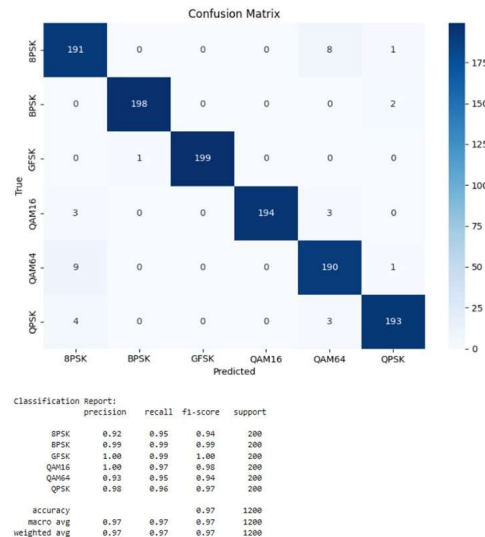


Figure 4.19: SNR = 4 dB

In Figure 4.18 model achieves excellent performance with 92% overall accuracy, and high precision, recall, and F1-scores across all classes. GFSK is perfectly classified, and QAM16/QAM64 confusion is greatly reduced compared to earlier models. The strong macro and weighted averages (0.92) indicate consistent and balanced classification across modulation types.

In Figure 4.19 model shows outstanding performance with 97% overall accuracy and nearly perfect classification across all modulation types. Confusion is minimal, especially for QAM16, QAM64, and GFSK, each with F1-scores of 0.94 to 1.00. High macro and weighted averages (0.97) confirm balanced and robust classification capabilities.

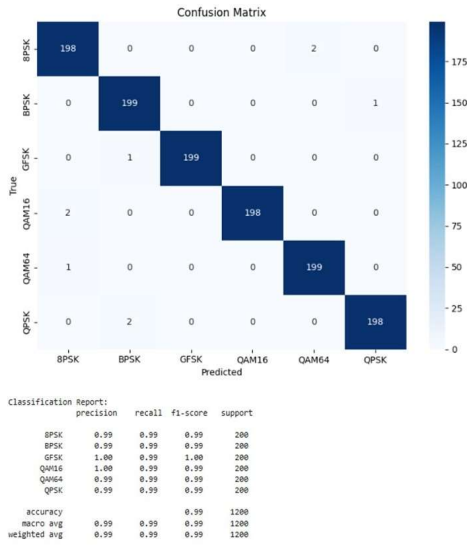


Figure 4.20: SNR = 8 dB

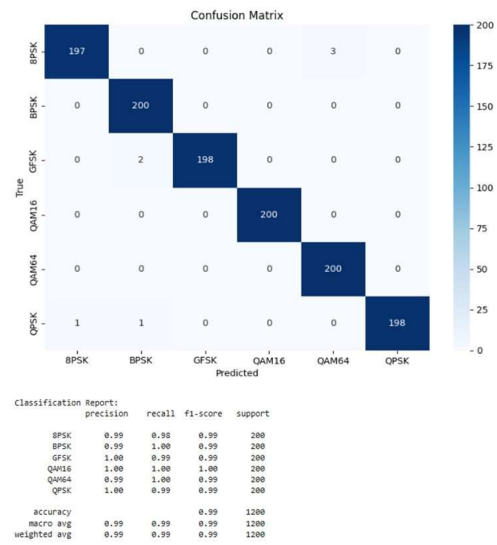


Figure 4.21: SNR = 12 dB

The Confusion Matrix in Figure 4.20 displays near-perfect classification across all modulation classes with an overall accuracy of 99%. All F1-scores range between 0.97 and 1.00, indicating excellent model consistency and performance. Very few misclassifications are present, confirming the model's high reliability and precision.

In Figure 4.21 confusion matrix shows very high classification accuracy (99%) across all signal types with almost no misclassifications. Precision, recall, and F1-scores are consistently at or near 0.99–1.00, reflecting outstanding model performance. The model demonstrates excellent reliability and generalization, making it well-suited for real-world modulation classification tasks.

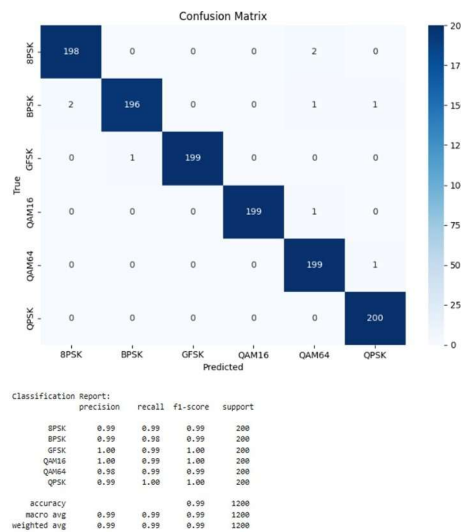


Figure 4.22: SNR = 16 dB

This confusion matrix in Figure 4.22 indicates exceptionally accurate classification, with only a few minor misclassifications among modulation types. Precision, recall, and F1-scores are above 0.98 for all classes, showing strong model consistency and balance. Overall accuracy is 99%, confirming a highly effective modulation recognition model.

For CNN model with the customized dataset, it is observed that as SNR increases from 0 to 16 dB with values of 0,4,8,12 and 16 dB the model gives an accuracy of 92%, 97%, 99%, 99% and 99% respectively thus averaging to a value of 97% approximately.

The results for CNN model shows that customized dataset gave better results with an average accuracy of 97% as against the average accuracy of 75% for the standard dataset.

The results for customized datasets for SVM and CNN show that the accuracy for SVM is 88% and for CNN it increases to 97%. Similarly the results for standard datasets for SVM and CNN shows approximately same the accuracy which are 76% and 75%.

Chapter 5

Conclusion

In this study, the modulation schemes GFSK, BPSK, QPSK, 8PSK, 64QAM, and 16QAM were successfully recognized using Support Vector Machine (SVM) and Convolutional Neural Network (CNN) models across varying Signal-to-Noise Ratios (SNR) of 0 dB, 4 dB, 8 dB, 12 dB, and 16 dB. The SVM model demonstrated a moderate performance on standardized datasets, achieving an accuracy range of 72% to 80%, while its performance improved significantly on customized datasets, reaching accuracies between 82% to 91%. On the other hand, the CNN model showed relatively lower accuracy on standardized datasets, ranging from 67% to 84%, but outperformed SVM on customized datasets, with a remarkable accuracy range of 92% to 99%. These results highlight the robustness of CNN in handling dataset-specific features and the consistent performance of SVM across various SNR levels, emphasizing the importance of dataset tailoring and model selection in Automatic Digital Modulation Recognition tasks.

Chapter 6

Future Scope

The future scope of Automatic Digital Modulation Recognition (ADMR) is highly promising, especially with the ongoing evolution of wireless communication systems. As we move toward 6G and beyond, the demand for intelligent and adaptive communication systems is growing. ADMR is expected to play a crucial role in these systems by enabling accurate and real-time identification of modulation schemes. This is vital for efficient spectrum utilization, interference mitigation, and secure communications. The integration of CNNs, known for their powerful feature extraction capabilities, with SVMs, which offer robust and fast classification, can lead to highly effective hybrid models suitable for deployment in real-world environments such as software-defined radios and cognitive radio networks.

In cognitive radio systems, the ability to quickly and accurately recognize modulation types allows for more dynamic and efficient spectrum access, making communication systems smarter and more responsive to changing environments. Furthermore, the future of ADMR may include the development of lightweight and energy-efficient models that can be deployed on edge devices, enabling modulation recognition in resource-constrained environments like IoT and mobile devices. Research may also focus on improving the robustness of ADMR systems against noise, fading, and adversarial attacks, ensuring reliability in harsh and unpredictable communication channels. While Automatic Digital Modulation Recognition (ADMR) is primarily used in fields like cognitive radio and spectrum monitoring, its potential application in healthcare is emerging, particularly in areas like tele-medicine and remote patient monitoring where reliable data transmission is crucial. For example, BPSK and QPSK used in body sensor network and medical telemetry, while QAM used in MRI and CT scan for data transmission. The combination of SVM and CNN will offer a solid foundation for building next-generation modulation recognition systems that are fast, accurate, and adaptable to future communication needs.

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PUBLICATION

Automatic Digital Modulation Recognition for Healthcare Using - SVM

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Abstract— In this paper, an Automatic Digital Modulation Recognition (ADMR) system is developed using Support Vector Machine (SVM). The system identifies modulation schemes such as GFSK, BPSK, QPSK, 8-PSK, 16-QAM, and 64-QAM from un-altered signal data at various Signal-to-Noise Ratios (SNR) from 0 to 16dB. These techniques find application in telemedicine and remote patient monitoring. Statistical features are extracted from the modulated signals to train the SVM model, which is refined using various kernel functions. The system performance is evaluated on both a standard dataset (RML2016.10a) and a newly developed custom dataset to analyze classification accuracy.

Index Terms— ADMR, ML, SVM, Modulation Schemes.

I. INTRODUCTION

In modern wireless communication systems, the ability to recognize and classify modulation schemes efficiently is crucial for signal intelligence, cognitive radio, and spectrum monitoring applications. This project focuses on Automatic Digital Modulation Recognition (ADMR) using SVM with an emphasis on extracting robust and discriminative features from received signals. The proposed system utilizes a feature engineering-driven approach by incorporating: Higher Order Moments (HOMs) and Higher-Order Cumulants (HOCs) to capture signal distribution characteristics. Frequency-domain features using FFT for spectral analysis. Wavelet Transform features to capture transient and multiresolution characteristics. Time-domain features such as skewness, kurtosis, and zero-crossing rate to characterize signal shape. Signal-to-Noise Ratio (SNR) impact analysis for robustness assessment. To enhance classification performance, Polynomial Feature Expansion and Feature Selection (PCA) were applied, followed by data augmentation using small Gaussian noise to improve model generalization. A Linear SVM classifier was optimized using Grid Search to determine the best hyper parameters for robust classification. The developed system successfully classifies six modulation schemes—GFSK, BPSK, QPSK, 8PSK, 16-QAM, and 64-QAM—at an SNR level of 0 dB, 4dB, 8dB, 12dB and 16dB. By combining handcrafted feature extraction with an optimized SVM classifier, the model achieves high accuracy while maintaining interpretability, making it a valuable tool for real-world communication applications where efficiency and reliability are crucial. While Automatic Modulation

Recognition (AMR) is primarily used in fields like cognitive radio and spectrum monitoring, its potential application in healthcare is emerging, particularly in areas like telemedicine and remote patient monitoring where reliable data transmission is crucial. For example, BPSK and QPSK used in body sensor network and medical telemetry, while QAM used in MRI and CT scan for data transmission.

II. LITERATURE SURVEY

Automatic Modulation Classification (AMC) has been a pivotal technique for identifying modulation formats without explicit overhead, thus enabling adaptive transmissions and cognitive radio systems. Likelihood-based methods, such as the tests introduced in [1], can be near-optimal under fully known channel parameters but tend to be computationally expensive and vulnerable to offsets or rapid channel variations. Accordingly, feature-based AMC has emerged as a more feasible alternative, extracting discriminative statistics—e.g., higher-order cumulants, wavelet transforms, or differential phase factors—and then classifying them via machine-learning models [2,3]. Empirical evidence in [4] and [5] shows that advanced feature engineering and SVM kernel optimization can significantly boost recognition under fading channels. In specialized scenarios such as biomedical implants, resource scarcity and severe path loss complicate AMC: Hannan et al. [7] surveyed low-power modulation schemes for in-body networks, whereas Oh et al. [8] demonstrated a high-data-rate implantable waveform requiring domain-specific classification logic. Similarly, Ko et al. [6] addressed robust watermarking for medical data, emphasizing the interplay between secure embedding and accurate signal handling. Telemedicine contexts, as discussed by Gayen et al. [9], impose additional real-time constraints on AMC due to unpredictable bandwidth and SNR conditions, while Iglesias et al. [10] underscored the immediate responsiveness needed in military broadband classification. While feature-based SVM pipelines remain practical, deep learning has gained prominence for its end-to-end capability and resilience to complex channel impairments. In [11], a CNN-GRU hybrid leverages temporal dependencies for MPSK, MFSK, and MQAM at low SNR, and further approaches incorporate specialized pre processing—

e.g., deep K-SVD denoising in [13] or resolution-transformed spectrograms in [14]. A Deep Belief Network in [12] similarly enhances recognition for ASK and PSK, and [15] demonstrates how CNNs can address phase offsets for OFDM transmissions. Collectively, these works affirm that a judicious blend of domain-specific feature extraction or deep-learning architectures, tailored to channel constraints, can achieve robust AMC performance across diverse

III. METHODOLOGY

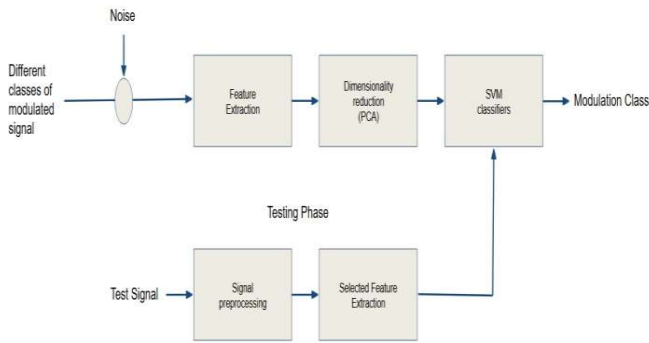


Figure 1. Proposed System

Figure 1 represents the outline of the proposed method. It consists of two main phases: training and testing. To analyze the performance, a set of modulation classes (1,000 samples of each class) was generated under varying conditions. For training, 80% of the dataset was used, while the remaining 20% was reserved for testing. The standard dataset was gathered from RML2016.10a, which contains various modulation types including GFSK, BPSK, QPSK, 8PSK, QAM16, and QAM64. Each signal sample was simulated under controlled conditions at an SNR ranging from 0 dB to 16 dB, to evaluate the classifier's performance under minimal noise. Additionally, a customized dataset was used, consisting of the same modulation types: GFSK, BPSK, QPSK, 8PSK, QAM16, and QAM64. These signal samples were also simulated under controlled SNR conditions (0 dB to 16 dB) in the presence of Additive White Gaussian Noise (AWGN). To capture the characteristics of the signals, both time-domain and frequency-domain features were extracted. These include statistical metrics such as skewness, kurtosis, energy, variance, and Zero Crossing Rate (ZCR), along with Hjorth parameters and wavelet-based features. Frequency-domain features included spectral centroid, spectral spread, spectral flatness, FFT mean, FFT max, and spectral entropy. Additionally, higher-order moments and cumulants (up to 12th order), higher-order ratios, Peak-to-Average Power Ratio (PAPR), and crest factor were computed to capture complex modulation patterns. These features served as inputs for the classifier

to distinguish between different modulation types. Label encoding was used to convert modulation types into a suitable format for classification. Feature normalization was applied using the Standard Scaler to ensure uniform scaling across all features. Polynomial feature expansion was introduced using the Polynomial Features technique to incorporate interaction terms and better capture non-linear relationships within the data. Principal Component Analysis (PCA) was applied to reduce the dimensionality of the feature set while retaining the most significant information. This step helped remove redundancy, improve model performance, speed up classification, and reduce overfitting. A Support Vector Machine (SVM) classifier with a linear kernel was employed to classify the modulation types. Hyperparameter tuning was conducted using Grid Search to identify the optimal regularization parameter C. The SVM model was trained on 80% of the dataset, with the remaining 20% used for testing. The model's performance was assessed using classification accuracy, which measures the proportion of correctly predicted instances. A confusion matrix was also used to evaluate misclassifications and identify prediction error patterns. The top-performing SVM model was saved for future reference.

I. SIMULATION AND RESULTS

Standard Dataset (RML2016.10a):



Figure 2. Confusion Matrix for SNR=0dB

In Figure [2], at 0 dB, QAM16, QAM64, and GFSK achieve high accuracy, while QPSK is often misclassified as 8PSK or BPSK, showing challenges with phase-based modulations.

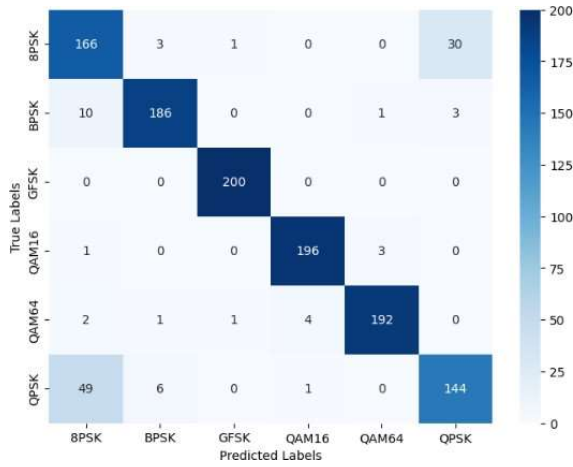


Figure 3. Confusion Matrix for SNR=4dB

In Figure [3] 4 dB, classification performance improves overall. GFSK, QAM16, and QAM64 show near-perfect results, and confusion among PSK schemes, especially QPSK, is notably reduced.



Figure 4. Confusion Matrix for SNR=8dB

In Figure [4] at 8 dB, the optimized Linear SVM achieves 90% accuracy, showing strong generalization. However, confusion between QPSK and 8PSK remains, even at higher SNR levels

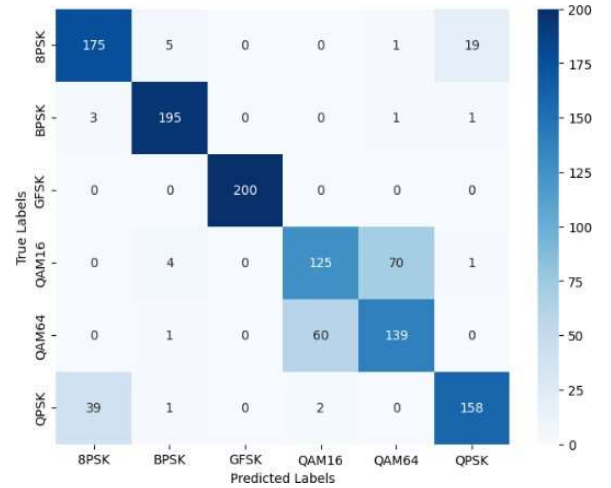


Figure 5. Confusion Matrix for SNR=12dB

In Figure [5] at 12 dB, overall accuracy drops slightly to 83%, with increased misclassification between QAM16 and QAM64, as well as 8PSK and QPSK.



Figure 6. Confusion Matrix for SNR=16dB

In Figure [6] at 16 dB, BPSK and GFSK are classified with high precision. However, QAM16 and QAM64 show frequent confusion, and some misclassification persists between 8PSK and QPSK due to their similar phase characteristics.

Customized Dataset:-

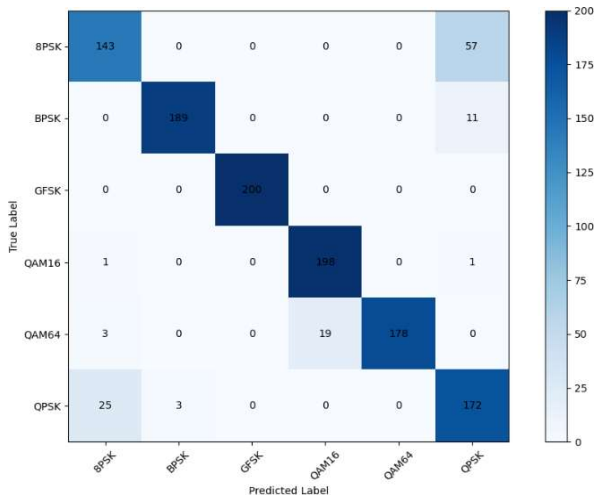


Figure 7. Confusion Matrix for SNR=0dB

In Figure [7] at 0 dB, the Linear SVM achieves 90% accuracy, with strong performance on QAM16, QAM64, and GFSK. However, QPSK is often confused with 8PSK and BPSK.

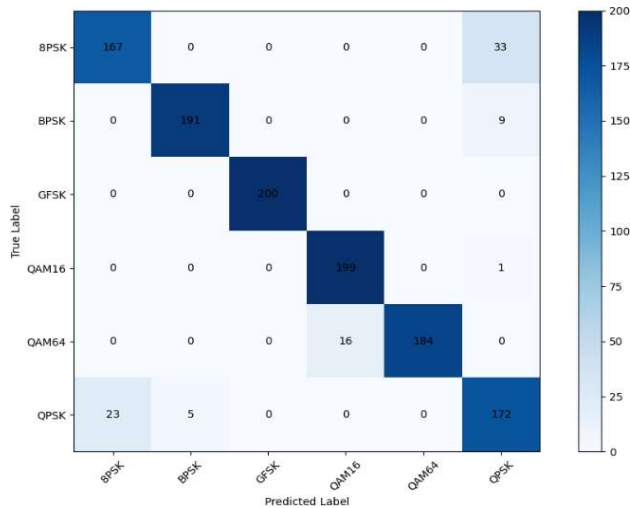
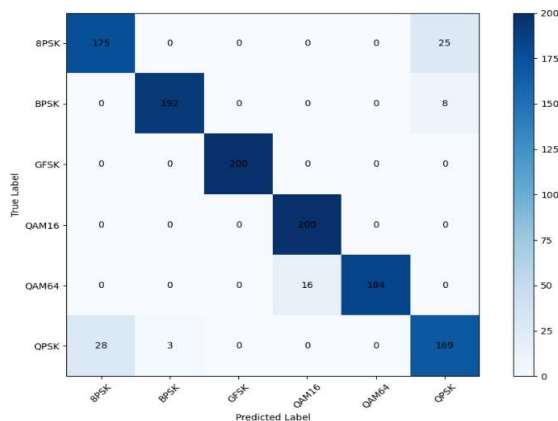


Figure 8. Confusion Matrix for SNR=4dB

In Figure [8] at 4 dB, accuracy improves to 92%. GFSK and QAM schemes are classified nearly perfectly, though some confusion between 8PSK and QPSK remains.



In Figure [9], at 8 dB, the Linear SVM achieves 93% accuracy, showing strong performance across most modulations. However, QPSK continues to be confused with 8PSK despite the higher SNR.

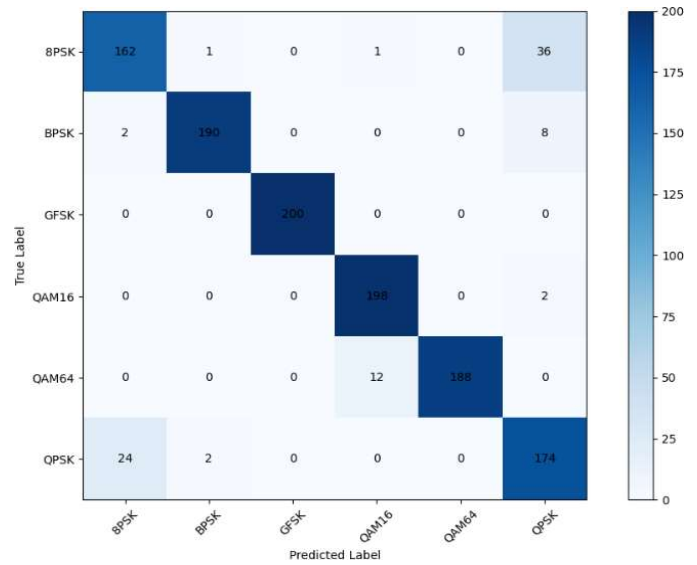


Figure 10. Confusion Matrix for SNR=12dB

In Figure [10] at 12 dB, accuracy remains at 93%. GFSK and QAM schemes are well classified, though some confusion persists between QAM16 and QAM64, as well as between QPSK and 8PSK.

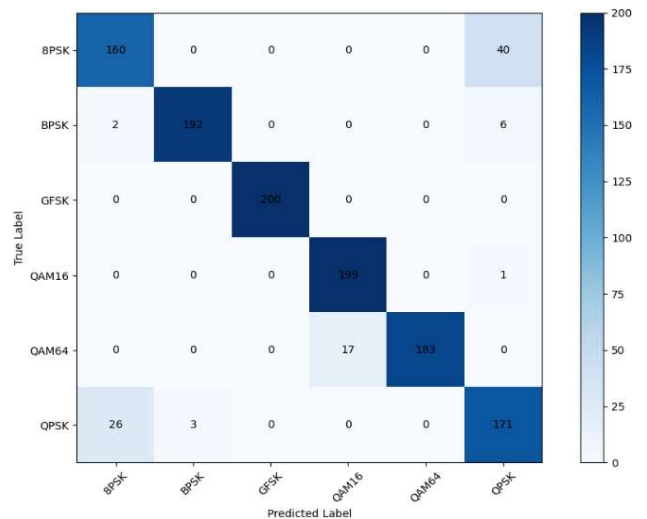


Figure 11. Confusion Matrix for SNR=16dB

In Figure [11], at 16 dB, the model maintains 93% accuracy. BPSK and GFSK are classified with high precision. Some confusion remains between QAM16 and QAM64, as well as between 8PSK and QPSK, due to their similar characteristics.

Table 2.Comparative Analysis of Accuracy for Standard and Customized Dataset

SNR (dB)	Standard Dataset Accuracy (%)	Proposed Dataset Accuracy (%)
0	85	90
4	90	92
8	90	93
12	83	93
16	75	93

The Customized dataset consistently outperforms the standard dataset across all SNR levels, demonstrating its effectiveness. The accuracy remains stable at 93% from SNR = 8 dB onwards, highlighting its robustness to noise. At lower SNR values, such as 0 and 4 dB, both datasets exhibit similar performance, though the proposed dataset shows a slight improvement. However, at higher SNR levels of 12 and 16 dB, the proposed dataset provides a significant performance boost over the standard dataset, further emphasizing its superiority in high-SNR conditions.

II. CONCLUSIONS

In this paper, an Automatic Digital Modulation Recognition (ADMR) system was developed using a SVM classifier. This system identifies six digital modulation schemes(GFSK, BPSK, QPSK,8 PSK,16-QAM, and 64-QAM from unaltered signal data at signal to noise ratios (SNR) ranging from 0 to 16dB. SVM model was trained using advanced statical features which were extracted from modulated signal . The performance of system was evaluated using standard dataset and a newly developed custom dataset to analyze classification accuracy. The simulation results show that the accuracy obtained using Standard Dataset was between 75-90%. The customized dataset gave better results where in the accuracy improved to 90-93%.

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